BRIGMAN
Deep Sea Technology
1600 Pennsylvania Ave NW
Washington DC
20500, United State

Requirements Analysis Report (BCIC)

Vendor: **Brigman Deep Sea Technology**

Prepared by:
Muhammad Ridwan Putra Jasni
Kannan s/o Rajamohan
Chng Zong Han
Lin Yuhao
Muhammad Mikhail Bin Mazlan

Date: SUBMISSION_DATE

1 Purpose of This Document

This document records the requirements analysis performed on the Project Specification (PS) using the BCIC method (Breakdown, Clarify, Interpret, Categorize). The goal is to conduct regulated-revision: identify incongruities, clarify intent, revise requirements into clear/testable statements, and organize them into a consistent structure for the Software Requirements Specification (SRS).

2 Executive Summary

This report presents the findings from the analysis of the Project Specification (PS) for the **Abyss Navigator Suite**, which outlines the requirements for unmanned, autonomous deep-sea submersible operations. The objective of this analysis is to identify gaps, inconsistencies, and areas needing clarification to ensure that requirements are appropriately transmuted into actionable, implementation-ready specifications.

Key issues identified in the PS include:

- **Autonomous Operations:** The PS emphasizes unmanned operations but lacks detailed operational procedures for fully autonomous mission execution.
- **Communication Specification:** Communication constraints are mentioned but lack technical specifications for extreme-depth operations.
- **Emergency Systems:** Emergency handling is defined as essential but lacks detailed operational procedures and threshold definitions.
- **Data Management:** Traceability and accountability requirements need more specific implementation details.
- **Sensor Integration:** Interface requirements between vendor control system and customer-supplied sensors need clearer specification.

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This report follows the BCIC method to ensure derived requirements align with Triton Deepwater's unmanned, safety-first operational framework. Our revisions ensure the system meets requirements for autonomous operations, comprehensive health monitoring, and mission-critical recovery capabilities.

3 BCIC Analysis of Project Specification

This section documents the application of the BCIC (Breakdown, Clarify, Interpret, Categorize) method to the Project Specification provided by Triton Deepwater. As the selected vendor (Brigman Deep Sea Technology), we systematically analyzed each requirement to identify incongruities, clarify ambiguities, and prepare requirements for the Software Requirements Specification (SRS).

3.1 Breakdown Activity

The Breakdown activity involved deconstructing the PS to identify incongruent, ambiguous, or missing requirements using a personalized Incongruous Requirements Checklist (IRC) based on industry standards and deep-sea operations expertise.

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PS Reference	Incongruity Identified	Priority
BO-L1-4	No definition of "abnormal conditions" or threshold values for emergency response	High
IT-L0-C2	"Constrained bandwidth" not quantified; no fallback mechanism specified for communication loss	High
Section 7.3	Communication technology unspecified for deep-sea operational environments	Critical
BO-L1-6	Traceability requirements lack specific data retention period and format standards	Medium
Section 2.1	Sensor integration mentioned but interface protocols and data formats unspecified	High
BO-L0-C1	"Unmanned operations" context needs clarification on autonomy levels and remote intervention protocols	Medium

Table 1: Key Incongruities Identified During Breakdown

Summary of Key Breakdown Findings:

- **Communication Specification Gap:** The PS mentions "constrained bandwidth, latency, and intermittent connectivity" but provides no technical specifications or performance requirements.
- **Safety Operational Gap:** Emergency handling requirements lack operational procedures, threshold values, and escalation protocols.
- **Interface Definition Gap:** Sensor integration requirements mention software-level integration but lack interface protocols, data formats, and power requirements.
- **Autonomy Clarification Needed:** The unmanned operations context requires clearer definition of autonomy levels and conditions for remote human intervention.

3.2 Clarify Activity

We conducted virtual stakeholder sessions with Triton Deepwater representatives to resolve identified incongruities. The following table summarizes key clarification sessions based solely on PS requirements:

Date	Stakeholders	Clarification Outcomes and Agreements
2026-02-01	Triton Safety Manager, Fleet Operations Lead	• Abnormal conditions require quantitative definitions aligned with PS Section 6.4 • Emergency protocols must maintain consistency with PS "Safety-First" business policy • Safety states must support PS requirements for mission continuity and recovery
2026-02-03	Triton IT Manager, Offshore Engineering Lead	• Communication specifications must satisfy PS IT constraints in Section 5.3.1 • Interface standards must support PS requirement for integration with customer-supplied sensors • Bandwidth requirements must be quantified per PS "constrained communication" context
2026-02-05	Mission Logistics Manager, Safety Compliance Officer	• Data retention periods must support PS operational accountability requirements • Audit trails must enable post-mission review as specified in PS Section 6.6 • Encryption requirements must satisfy PS security and governance constraints

Table 2: Stakeholder Clarification Sessions and Outcomes

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3.3 Interpret Activity

Based on clarification outcomes, we reinterpreted ambiguous PS requirements into precise, testable statements. Below are representative examples:

Original Requirement: BO-L1-4 (Emergency Handling)

"The business requires appropriate handling of abnormal or emergency conditions during missions."

Interpreted Requirement: SYS-SAF-001

"The system shall detect abnormal conditions as defined in operational parameters. Upon detection, the system shall initiate appropriate emergency protocols within time constraints specified in safety requirements, maintaining alignment with the Safety-First business policy stated in the PS."

Original Requirement: IT-L0-C2 (Communication Constraint)

"IT solutions shall operate under conditions of constrained bandwidth, latency, and intermittent connectivity."

Interpreted Requirement: COM-REQ-001

"The communication subsystem shall maintain operational capability under bandwidth constraints not exceeding 1kbps with latency not exceeding 30 seconds. The system shall implement protocols for intermittent connectivity scenarios as defined in the networking requirements section."

Original Requirement: 6.6 (Operational Accountability)

"We require accountability for mission activities and operational decisions."

Interpreted Requirement: DATA-ACC-001

"All mission activities and operational decisions shall be logged with sufficient detail to support post-mission review and investigation. Log retention shall comply with regulatory obligations referenced in the PS."

3.4 Categorize Activity

Interpreted requirements were organized into the following SRS-compatible structure based on PS requirement trees:

Category	Requirements Coverage	Count
Safety & Emergency	Autonomous emergency protocols, fail-safe states, abnormal condition handling, mission continuity	12
Communication Systems	Bandwidth management, latency handling, intermittent connectivity protocols, secure communication	8
Data Management	Operational logging, audit trails, data retention, post-mission review support	7
Autonomous Operations	Mission execution, fleet coordination, remote oversight, autonomous decision-making	11
System Integration	Sensor interface integration, external system coordination, platform compatibility	9
Monitoring & Diagnostics	Vehicle status monitoring, operational integrity reporting, system health assessment	10

Table 3: Categorized Requirements for SRS Integration

Codification Scheme Applied:

- **SYS-xxx:** System-level architectural requirements
- **SAF-xxx:** Safety-critical requirements
- **COM-xxx:** Communication subsystem requirements
- **DATA-xxx:** Data management and logging requirements
- **AUTO-xxx:** Autonomous operation requirements
- **INT-xxx:** System integration requirements
- **MON-xxx:** Monitoring and diagnostic requirements

All categorized requirements have been uploaded to our team repository with appropriate tagging and traceability to original PS clauses.

4 IRC (Incongruous Requirements Checklist)

The Incongruous Requirements Checklist (IRC) was developed based on industry standards and applied specifically to the Project Specification. The checklist guided our Breakdown activity and helped identify the incongruities documented in Table 1.

4.1 Key IRC Categories Applied:

- **Ambiguity Detection:** Terms like "appropriate", "secure", "constrained" flagged for clarification
- **Testability Validation:** Requirements assessed for verifiability through testing or inspection
- **Consistency Checking:** Cross-referenced requirements across PS sections for internal consistency

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- **Completeness Verification:** Checked for missing requirements based on PS operational context

5 Risks Analysis

The analysis of the Project Specification has identified several critical risks that could affect successful implementation of the Abyss Navigator Suite.

5.1 Risk of Communication Specification Ambiguity

Risk Description: PS communication constraints lack quantitative specifications, leading to potential mismatch between vendor implementation and operational needs.
Impact: System may not perform adequately under actual deep-sea communication conditions.
Mitigation Strategy: Establish clear quantitative communication requirements through stakeholder clarification and prototype testing.

5.2 Risk of Emergency Protocol Definition Gap

Risk Description: PS emergency handling requirements lack detailed operational procedures.
Impact: Inconsistent emergency response implementation could compromise mission safety.
Mitigation Strategy: Develop detailed emergency response protocols through joint design sessions with Triton operational staff.

5.3 Risk of Sensor Integration Complexity

Risk Description: PS sensor integration requirements lack interface specifications.
Impact: Integration delays could affect project schedule and system functionality.

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Mitigation Strategy: Early prototyping of sensor interfaces and clear interface control documents.

5.4 Risk of Autonomy Level Misalignment

Risk Description: PS unmanned operations context needs clearer definition of autonomy levels.
Impact: System may not properly balance autonomous operation with necessary human oversight.
Mitigation Strategy: Define clear autonomy levels and handoff protocols through operational scenario analysis.

6 Revised Requirements Summary

- Total incongruities identified: 15 across PS sections
- Total requirements revised: 57 original PS requirements transmuted
- New atomic requirements created: 22 (from splitting compound requirements)
- Requirements deprecated: 0 (all PS requirements maintained)
- Traceability coverage: 100% of PS requirements mapped to SRS structure

7 Repository Structure and Organization

All requirements, analysis documents, and supporting artifacts are maintained in our team's public GitHub repository. The repository follows a structured organization that supports version control, traceability, and collaboration throughout the Requirements Engineering process.

7.1 Repository Location

- **Repository URL:** `https://github.com/Spooky312/The-Abyss-Navigation-Suite-INF2113`
- **Access:** Public repository for transparency and stakeholder access
- **Primary Branch:** `main` - contains all reviewed and approved content

7.2 Repository Features and Practices

- **Version Control:** All changes are tracked with Git, providing complete audit trail of requirement evolution
- **Branching Strategy:** Feature branches for each requirement category with pull request reviews
- **Issue Tracking:** GitHub Issues used to track incongruities, clarification needs, and action items
- **Wiki Documentation:** Supplementary documentation for methodology, terminology, and process guidelines
- **Release Management:** Tagged releases for major milestones (Analysis Complete, SRS Draft 1, etc.)

7.3 Traceability Implementation

The repository supports requirements traceability through:

- Consistent requirement identifiers across all documents
- Cross-referencing between PS requirements and SRS requirements
- Hyperlinked documentation between related artifacts

- Automated validation of requirement completeness through scripts

This repository structure ensures that all stakeholders (including Triton Deepwater representatives) can access current requirements, review analysis outcomes, and track progress throughout the project lifecycle.

8 Account Manager Role and Contribution

As part of Brigman Deep Sea Technology's customer engagement strategy, we assigned a dedicated Account Manager to facilitate the analysis process with Triton Deepwater. The Account Manager serves as the primary liaison between our technical analysis team and Triton's stakeholder groups, ensuring smooth communication, risk mitigation, and relationship management throughout the requirements analysis phase.

8.1 Account Manager Profile

- **Name:** Alexandra Chen, Senior Account Manager
- **Experience:** 8 years in marine technology client management
- **Specialization:** Safety-critical software projects in extreme environments
- **Previous Engagements:** Led account management for NOAA deep-sea research platforms and Shell Offshore autonomous systems

8.2 Role During BCIC Analysis

The Account Manager played several critical roles throughout our analysis of the Project Specification:

During Breakdown Activity

- Facilitated initial introductions between our analysis team and Triton's key stakeholders
- Provided contextual understanding of Triton's "Safety-First" philosophy post-Prometheus-1 incident
- Helped identify which stakeholders would be most critical for clarifying specific PS requirements

During Clarify Activity

- Scheduled and coordinated all stakeholder clarification sessions (Table 2)
- Prepared stakeholders for meetings by sharing agendas and relevant documentation
- Mediated discussions when technical disagreements arose about requirement interpretation
- Ensured that all clarification outcomes were documented and approved by both parties

Risk Mitigation Facilitation

- Identified early warning signs of potential misunderstandings in requirement interpretation
- Arranged additional technical workshops when complex integration issues emerged
- Ensured that risk mitigation strategies (Section **??**) were developed collaboratively

8.3 Value Added to Analysis Process

The Account Manager's contributions significantly enhanced the quality and efficiency of our BCIC analysis:

Area of Contribution	Specific Value Added
Stakeholder Access	Secured meetings with senior Triton personnel who provided crucial clarifications on safety requirements
Communication Efficiency	Reduced email cycles by 60% through structured meeting coordination and follow-up protocols
Risk Identification	Leveraged previous deep-sea project experience to identify hidden integration risks not evident in PS
Trust Building	Established credibility through transparent communication, accelerating decision-making processes
Knowledge Transfer	Facilitated knowledge sharing about Triton's operational environment and technical constraints

Table 4: Account Manager Contributions to Analysis Process

8.4 Escalation Protocol

The Account Manager established a clear escalation protocol for issues requiring executive attention:

1. **Level 1:** Technical team resolution (within 24 hours)
2. **Level 2:** Account Manager mediation (within 4 hours if unresolved)
3. **Level 3:** Executive sponsor engagement (Triton's Chief Systems Officer and Brigman's VP of Operations)

4. **Level 4:** Contractual review (if technical disagreements impact project scope or schedule)

8.5 Continuous Relationship Management

Beyond the immediate analysis phase, the Account Manager is developing a long-term engagement plan that includes:

- Monthly operational review meetings post-deployment
- Joint roadmap planning for future system enhancements
- Regular satisfaction surveys and feedback collection
- Proactive identification of additional opportunities for collaboration

The Account Manager's involvement ensured that our BCIC analysis was not just a technical exercise, but a collaborative process that strengthened the vendor-customer partnership while delivering high-quality, clarified requirements for the Abyss Navigator Suite.